

Week 6 Recitation

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Seller's Problem

The goal and the 3 ingredients

Goal of the **Buyer**: Maximize **Utility** (Given the budget)

Goal of the **Seller**: Maximize **Profit** = Total Revenue - Total Cost

Recall: We are still analyzing **Perfectly competitive market!**

- So many buyers and sellers – Individuals are price takers
- Homogeneous products
- Free entry and exit in the long run

Seller's Problem

The goal and the 3 ingredients

Goal of the **Seller**: Maximize **Profit** = Total Revenue - Total Cost

3 Ingredients of Seller's problem

1. How can I
make the product?
(=**Production**:
turning inputs into
outputs!)

2. What is the cost of
making the product?
(=**Cost**)

Seller's Problem

The goal and the 3 ingredients

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3 Ingredients of Seller's problem

1. How can I make the product?
(=**Production**: turning inputs into outputs!)

- Inputs(Factors), Outputs
 - Short run / Long run
 - Fixed factor of production
 - Variable factor of production
 - Marginal Product
-

2. What is the cost of making the product?
(=**Cost**)

Seller's Problem

The goal and the 3 ingredients

Goal of the **Seller**: Maximize **Profit** = Total Revenue - Total Cost

3 Ingredients of Seller's problem

1. How can I make the product?
(=**Production**: turning inputs into outputs!)

- Inputs(Factors), Outputs
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2. What is the cost of making the product?
(=**Cost**)

- Total Cost = Variable Cost + Fixed Cost (Short-run)
- $\frac{\text{Total Cost}}{Q} = \frac{\text{Variable Cost}}{Q} + \frac{\text{Fixed Cost}}{Q}$
- Marginal Cost (MC) = $\frac{\Delta \text{Total Cost}}{\Delta Q}$
 $= \frac{\Delta VC}{\Delta Q}$

Seller's Problem

3. How much money can the seller get for the product in the market (= **Revenue**)

- Total Revenue (TR)
= Price (P) $\times$ Quantity Sold (Q)
 - Marginal Revenue (MR)
= $\frac{\Delta \text{Total Revenue}}{\Delta Q} = P$
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Optimal Q^* happens at:

Marginal Revenue (MR) = Marginal Cost (MC).

Seller's Problem

1. Production

Marginal product: how much additional output is produced by employing one more input.

- Marginal Product (MP): The change in total output employing an additional input unit

$$MP_i = \frac{\Delta q}{\Delta i}$$

- q = output
- i = a factor of production
- Marginal product *will eventually* exhibit **diminishing returns** (MC $\uparrow$) from capacity constraints
- Marginal product *can* exhibit increasing returns through specialization (MC $\downarrow$)

Seller's Problem

1. Production

Example: Production of Cheese Boxes

# of workers	Total Output	Marginal Product (MP_L)
0	0	—
1	100	$\frac{100-0}{1-0} = 100$
2	207	$\frac{207-100}{2-1} = 107$
3	321	$\frac{321-207}{3-2} = 114$
4	444	$\frac{444-321}{4-3} = 123$
5	558	$\frac{558-444}{5-4} = 114$
6	664	$\frac{664-558}{6-5} = 106$
7	762	$\frac{762-664}{7-6} = 98$

- Labor is the only variable factor of production
- Over the range: 0 - 4 workers,
 - production exhibits **increasing returns to labor**
 - MC will be decreasing over this range of output produced
- Over the range: 5 - ∞ workers,
 - production exhibits **decreasing returns to labor**
 - MC will be increasing over this range of output produced

Seller's Problem

2. Cost

Short-run vs. Long-run

Example: Determine if this firm is operating in the short-run or long-run. What are the variable and fixed cost components (if they exist) of the total cost equation?

$$TC = 100 + \frac{1}{2}q^2$$

Short-run or long-run? **Short-run**, how do we know?

Seller's Problem

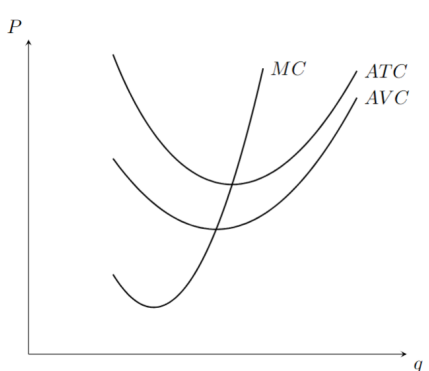
2. Cost

Total output (q), fixed costs, and variable costs are given here.
Do the calculations on your own for $L = 2, 3, 4, 5$.

L	q	MP_L	FC	VC	TC	AFC	AVC	ATC	MC
0	0	–	10	0	10	–	–	–	–
1	6	$\frac{6-0}{1-0} = 6$	10	36	$10 + 36 = 46$	$\frac{10}{6} = 1.67$	$\frac{36}{6} = 6$	$\frac{46}{6} = 7.67$	$\frac{46-10}{6-0} = 6$
2	18	12	10	72	82	0.56	4	4.56	3
3	27	9	10	108	118	0.37	4	4.37	4
4	33	6	10	144	154	0.30	4.36	4.67	6
5	36	3	10	180	190	0.28	5	5.28	12

Seller's Problem

2. Cost



- MC intersects ATC and AVC at their minimum points
- ATC and AVC asymptotically converge as q becomes larger due to AFC converging to zero
- ATC (or AVC) decreases when MC is less than ATC (or AVC).

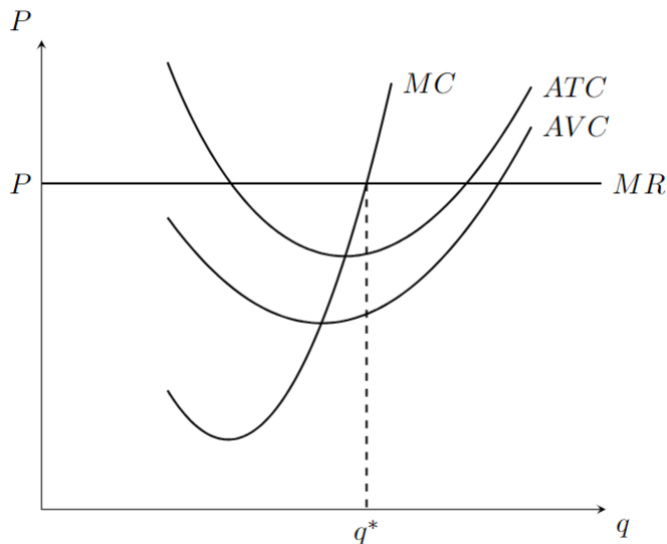
Seller's Problem

3. Revenue

- The **benefit** firms(sellers) receive from selling the goods is known as revenue!
- **Marginal revenue** (MR): The additional revenue they get for selling one more unit of output
- MR is the same thing as MB to the firm(the seller)
- In the competitive market, the seller is price-taker, so the price (P) does not depend on Q (i.e., P is constant)
 $\Rightarrow MR = P$ is constant!

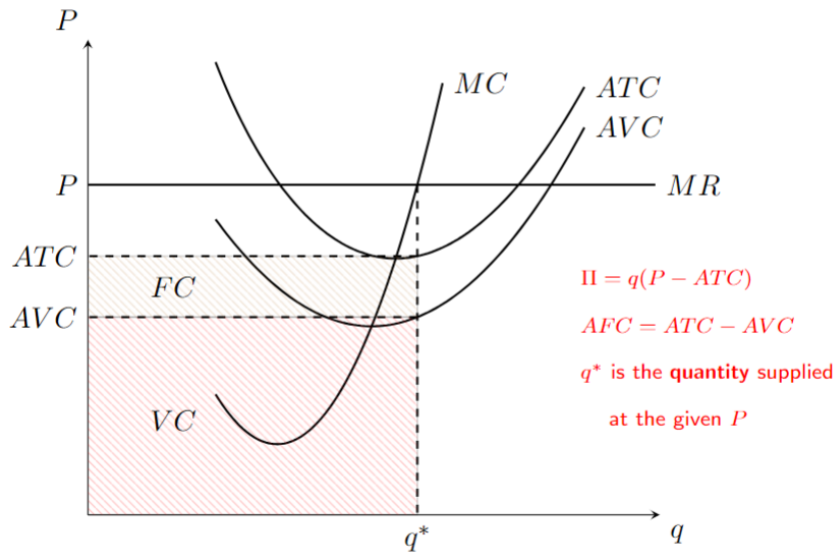
Seller's Problem

Seller's Decision - Optimal Q-supplied and Profit



Seller's Problem

Seller's Decision - Optimal Q-supplied and Profit



Seller's Problem

Seller's Decision - Shut-down

Shutdown is also a **short-run** decision.

- The firm is still operating as a business, but is choosing not to produce any output
- Because of fixed factors of production in the short-run, firms are not able to exit the market

If $\Pi = -FC$, the firm is indifferent between shutting down production and producing q

If $\Pi < -FC$, the firm maximizes its profit (minimizing the loss) by shutting down production

The shutdown point is where $P = MC = AVC$
: The minimum point of the AVC curve

Seller's Problem

Seller's Decision - Shut-down

Why is that? (Just for fun)

We can express the profit (Π) as

$$\begin{aligned}\Pi &= TR - TC \\ &= P \cdot Q - ATC \cdot Q \\ &= Q(P - ATC)\end{aligned}$$

Then we can express the shutdown threshold as

$$\begin{aligned}\Pi &= -FC \\ \Leftrightarrow Q(P - ATC) &= -Q \cdot AFC \\ \Leftrightarrow P - ATC &= -AFC \\ \Leftrightarrow P &= ATC - AFC \\ \Leftrightarrow P &= AVC\end{aligned}$$